A COMPLETE DEVISSAGE PERSISTS UNDER AN AFFINE BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros; no macro was invented or redefined.

Editorial note. Same-chapter referenced results are located at their \label line inside the pinned chapter file; the cross-chapter affine base change lemma is located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim line carrying its canonical tag.

Allowed standard prerequisites (editorial).

- Complete devissage of a finite type module. Same chapter, Definition environment labelled definition-complete-devissage, source line 1195 (Stacks tag 05HH). Supplies the tuple (Z_k, Y_k, i_k, pi_k, G_k, alpha_k) and condition (2) of the definition, which is exactly what the counted proof has to preserve under base change.
- Base change of a one step devissage. Same chapter, Lemma labelled lemma-base-change-one-step, source line 879 (Stacks tag 05H7). Supplies the one-step half of the conclusion, quoted at proof lines 1288-1290; the remaining cokernel and condition-(2) arguments stay inside the counted span.

Source reference. Original definition 05HH: sources/stacks-flat/pinned-original-source.tex, source lines 1195–1195. Same-chapter pointer to the definition quoted both in the statement (line 1273) and in the proof (line 1294) for condition (2); the verification that this condition survives base change is the counted obligation and stays in the delivered body.

Source reference. Original lemma 05H7: sources/stacks-flat/pinned-original-source.tex, source lines 879–879. Same-chapter pointer used at proof line 1289 for the one-step half of the conclusion; the cokernel and condition-(2) arguments that complete the proof are retained verbatim in the counted span.

Source reference. Original lemma 02KG: sources/stacks-chow/tags.txt, source lines 3371–3371. Cross-chapter standard prerequisite invoked once at proof line 1298 to identify pi'_{k,*}G'_k with the pullback of pi_{k,*}G_k; it supplies that identification only, while the cokernel computation and the condition-(2) computation are retained in the counted span.

Lemma 0.1. *Let S, X, $\mathcal{F}$, s be as in Definition 05HH. Let $(S', s') \to (S, s)$ be any morphism of pointed schemes. Let $(Z_k, Y_k, i_k, \pi_k, \mathcal{G}_k, \alpha_k)_{k=1,\ldots,n}$ be a complete dévissage of $\mathcal{F}/X/S$ over s. Given this data let $X', Z'_k, Y'_k, i'_k, \pi'_k$ be the base changes*

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of X, Z_k, Y_k, i_k, π_k via $S' \to S$. Let $\mathcal{F}'$ be the pullback of $\mathcal{F}$ to X' and let $\mathcal{G}'_k$ be the pullback of $\mathcal{G}_k$ to Z'_k. Let α'_k be the pullback of α_k to Y'_k. If S' is affine, then $(Z'_k, Y'_k, i'_k, \pi'_k, \mathcal{G}'_k, \alpha'_k)_{k=1,\ldots,n}$ is a complete dévissage of $\mathcal{F}'/X'/S'$ over s'.

Proof. By Lemma 05H7 we know that the base change of a one step dévissage is a one step dévissage. Hence it suffices to prove that formation of $\mathrm{Coker}(\alpha_k)$ commutes with base change and that condition (2) of Definition 05HH is preserved by base change. The first is true as $\pi'^{*}_{k,*}\mathcal{G}'_k$ is the pullback of $\pi_{k,*}\mathcal{G}_k$ (by Cohomology of Schemes, Lemma 02KG) and because $\otimes$ is right exact. The second because by the same token we have

$$(\pi_{k,*}\mathcal{G}_k)_{\xi_k} \otimes_{\mathcal{O}_{Y_k,\xi_k}} \kappa(\xi_k) \otimes_{\kappa(\xi_k)} \kappa(\xi'_k) \cong (\pi'_{k,*}\mathcal{G}'_k)_{\xi'_k} \otimes_{\mathcal{O}_{Y'_k,\xi'_k}} \kappa(\xi'_k)$$

with obvious notation. □